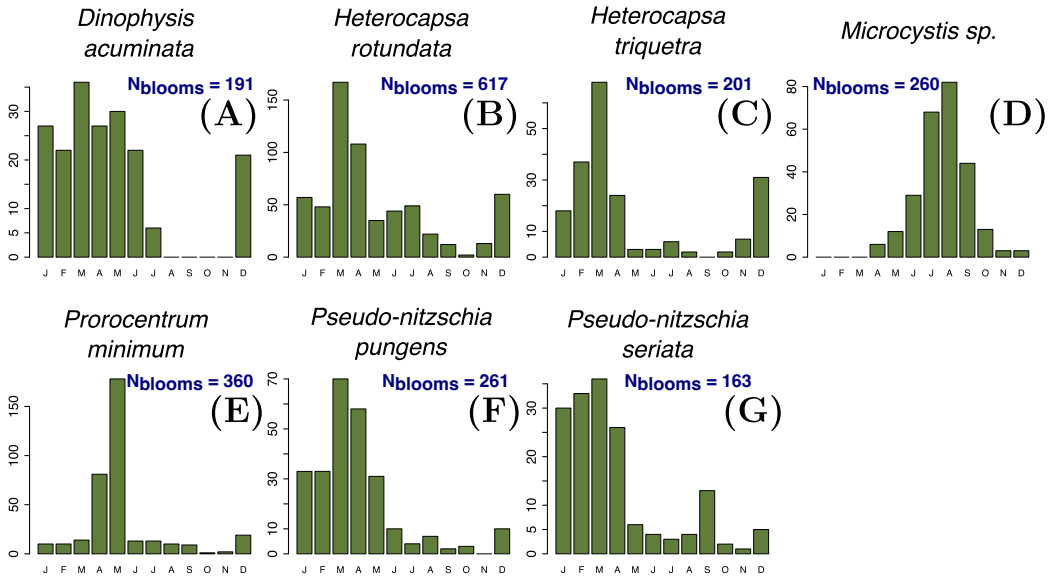
Supplementary Material

Supplementary figures

The temporal and spatial distribution of the observed algal blooms are shown in Figs. S1-S2, respectively. The comparison of the mechanistic model outputs with observational data appear in Fig. S3. Figure S4 shows the Pearson correlation between the environmental variables. The associations between environmental variables and algal bloom probability, and model skill coming from the regression tree analysis, are shown in Figs. S5-S6, respectively. The optimal variable combination results from selecting the five variables with the highest averaged absolute Shapley values. An association between environmental variables and algal bloom probability is quantified by first computing the Pearson correlation γ_i^k between environmental variable x_i and the Shapley values using training data subset k. Next, similar to the procedure for the GLMs, we train 10 regression trees since the random selection of a training and test data set affects the estimation of the γ-values. We compute the averaged γ-value signs of these 10 regression trees for each environmental variable x_i:

$$\gamma_i^{\pm} = \frac{1}{10} \sum_{k=1}^{10} \text{sign}(\gamma_i^k), \tag{S1}$$

To lower the computational cost, we trained 10 instead of 50 regression tree models for each variable combination and used the *fastshap* (0.1.1) package within the R software (version 4.2.1; 2022-06-23) to compute the Shapley values (Štrumbelj and Kononenko, 2014). The regression trees were generated using the *rpart* (4.1.16) package.



† N_{bloom} represents the number of blooms.

Figure S1: Temporal distribution (Jan-Dec) of the number of blooms reported in the Chesapeake Bay Program dataset of 1985-2020 for **(A)** *Dinophysis acuminata*, **(B)***Heterocapsa rotundata*, **(C)** *Heterocapsa triquetra* (or *steinii*), **(D)** *Microcystis sp.*, **(E)** *Prorocentrum minimum* (also known as *P. cordatum*), **(F)** *Pseudo-nitzschia pungens*, and **(G)** *Pseudo-nitzschia seriata*. Copied from Horemans et al. (2024) under the CC BY 4.0 license.

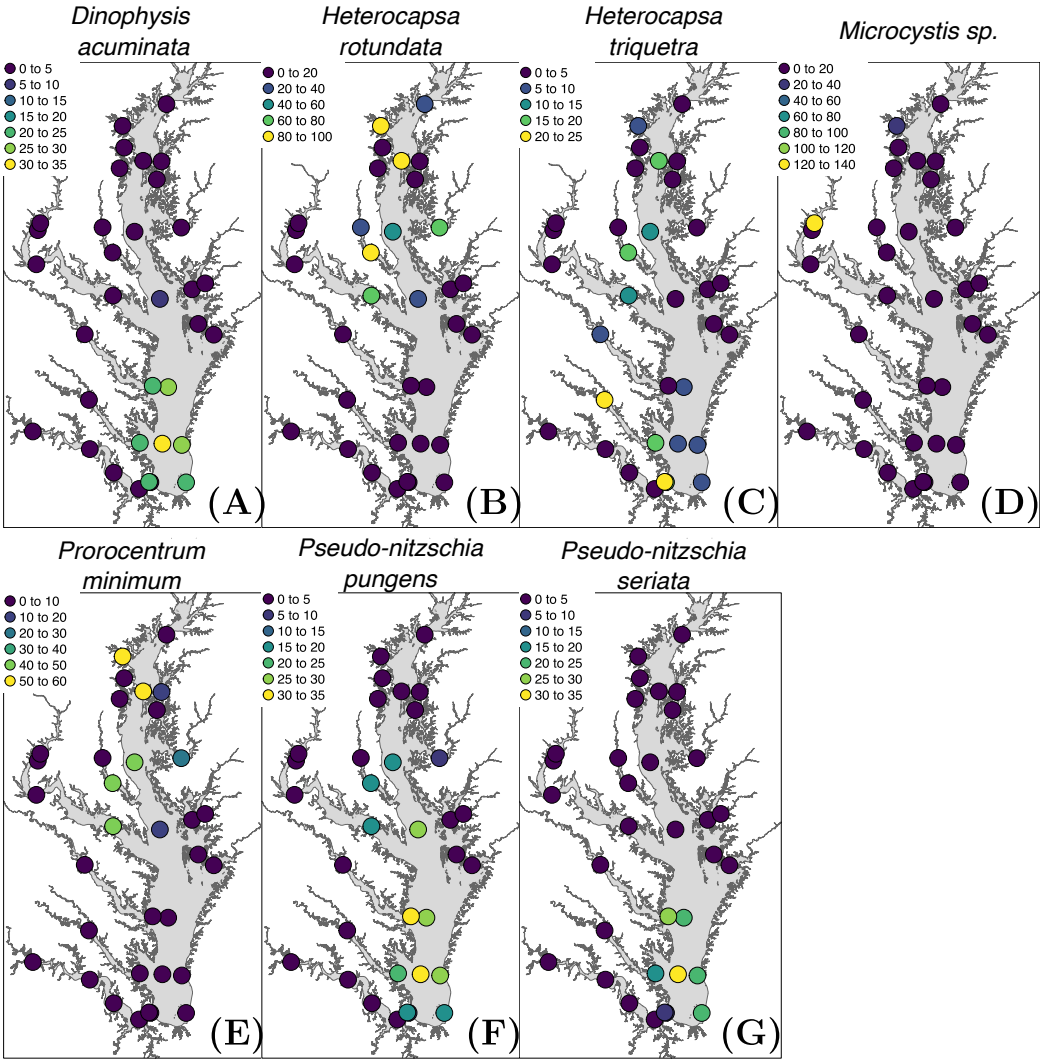
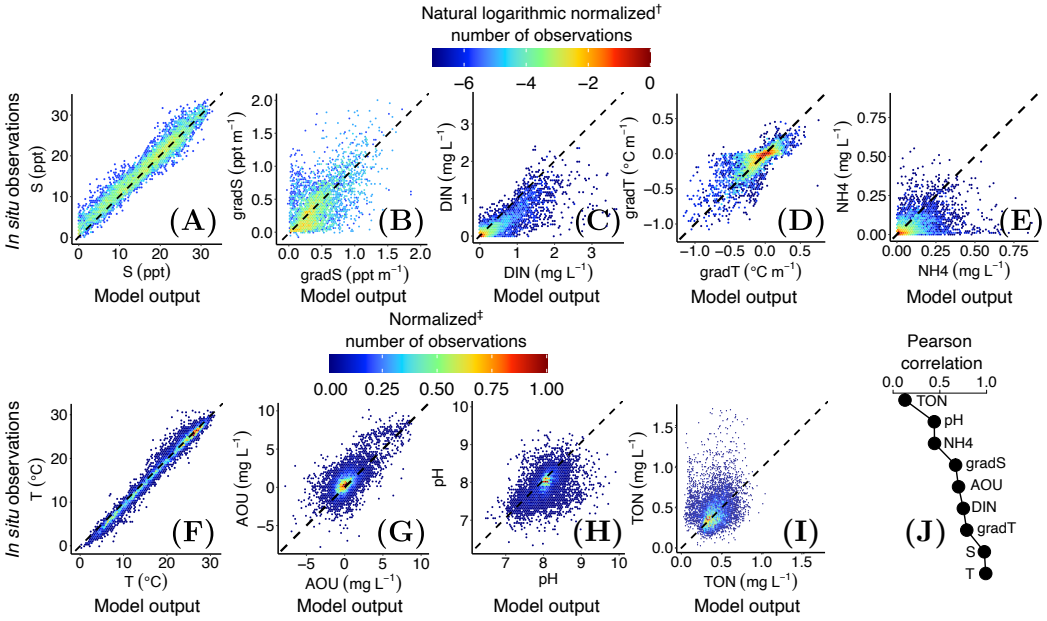


Figure S2: Spatial distribution of the number of blooms reported in the Chesapeake Bay Program dataset of 1985-2020 for **(A)** *Dinophysis acuminata*, **(B)***Heterocapsa rotundata*, **(C)** *Heterocapsa triquetra* (or *steinii*), **(D)** *Microcystis sp.*, **(E)** *Prorocentrum minimum* (also known as *P. cordatum*), **(F)** *Pseudo-nitzschia pungens*, and **(G)** *Pseudo-nitzschia seriata*. Copied from Horemans et al. (2024) under the CC BY 4.0 license.



† The normalization factors are 306, 284, 1,014, 175, and 606, respectively. We show a logarithmic scale because many observations are clustered near the origin of the figures.

‡ The normalization factors are 53, 80, 114, and 25, respectively.

Figure S3: Environmental mechanistic model output to *in situ* observations plot and normalized number of data points (colors) for **(A)** salinity (S), **(B)** vertical gradient in salinity (gradS), **(C)** dissolved inorganic nitrogen (DIN), **(D)** vertical gradient in water temperature (gradT), **(E)** ammonium (NH4), **(F)** water temperature (T), **(G)** apparent oxygen utilization (AOU), **(H)** pH, **(I)** total organic nitrogen (TON), and **(J)** the corresponding Pearson correlation coefficients. The total number of observations (i.e., 7,059) is the same for all subfigures. Adapted from Horemans et al. (2024) under the CC BY 4.0 license (gradT and NH4 plots were added).

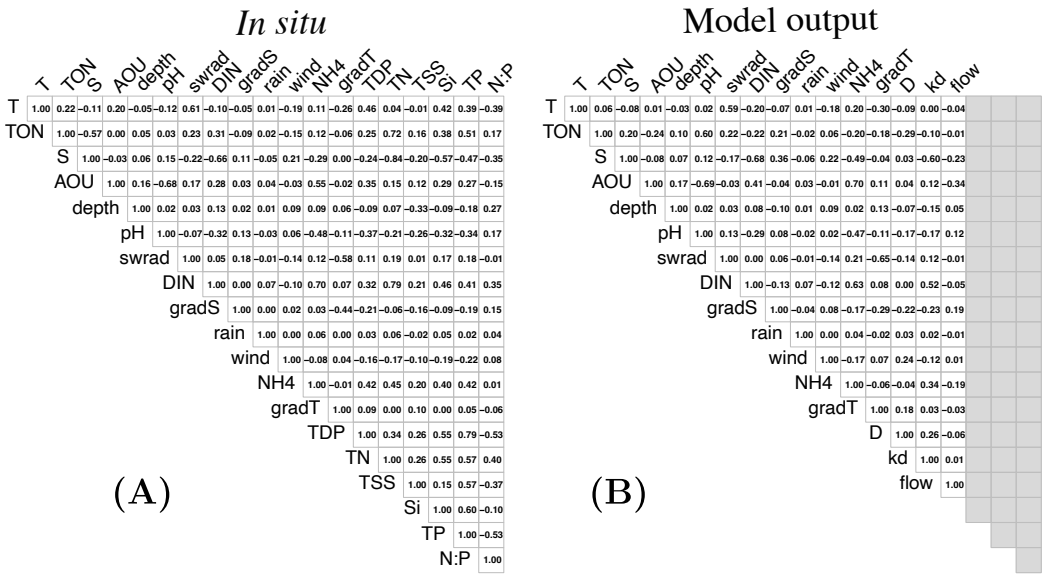


Figure S4: Collinearity between the environmental variables included in the A) *in situ* observations and B) mechanistic model output quantified by the Pearson correlation. The definition of the acronyms are listed in Table 2.

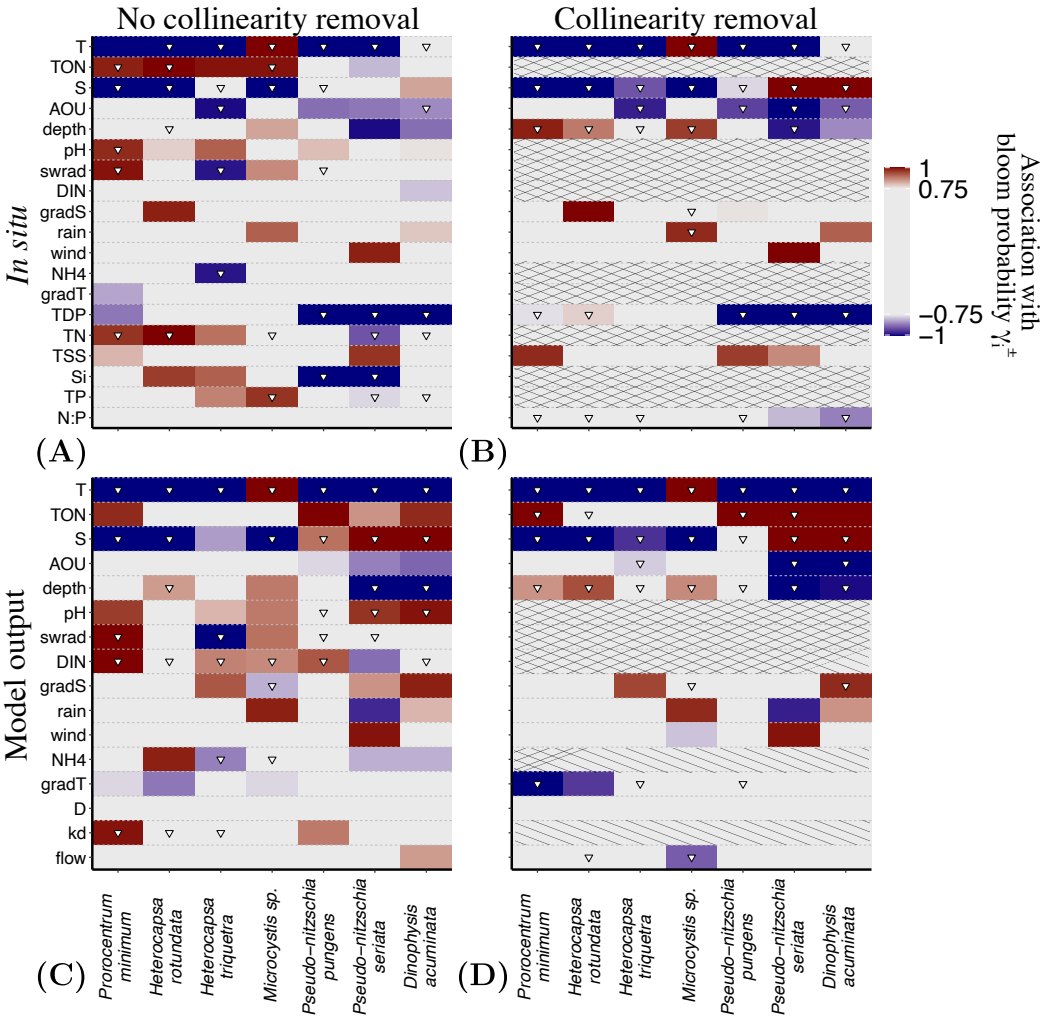


Figure S5: Associations between environmental variables and algal bloom probability [see Eq. (S1)], and the optimal variable combinations to forecast blooms (triangles) based on regression tree models for the seven algal taxa examined without (cf., correlation threshold of 1; left panels) and with (right panels) removal of collinearity using environmental (A)-(B) *in situ* observations and (C)-(D) mechanistic model output. The hashed rows depict the environmental variables that were removed based on a correlation threshold of 0.4. The definition of the acronyms are listed in Table 2.

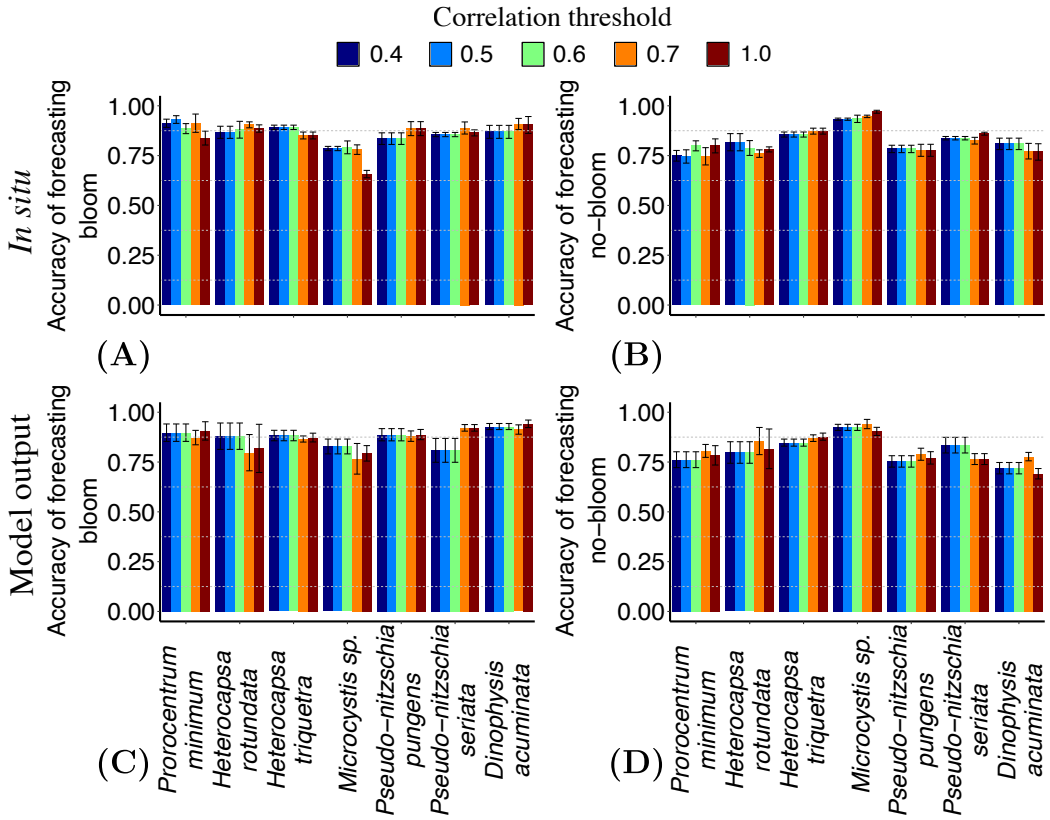


Figure S6: Regression tree model accuracy of forecasting a bloom and no-bloom occurrence for the seven algal taxa examined using various correlation thresholds (colored symbols) in which the regression tree model is trained using environmental (A)-(B) *in situ* observations and (C)-(D) mechanistic model output. The colored vertical lines depict the standard deviations of the accuracies of the 10 regression tree models that were constructed for each optimal set of environmental predictors.

References

Horemans, D.M.L., Friedrichs, M.A.M., St-Laurent, P., Hood, R.R., Brown, C.W., 2024. Evaluating the skill of correlative species distribution models trained with mechanistic model output. Ecological Modelling 491, 110692. doi:10.1016/J.ECOLMODEL.2024.110692.

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